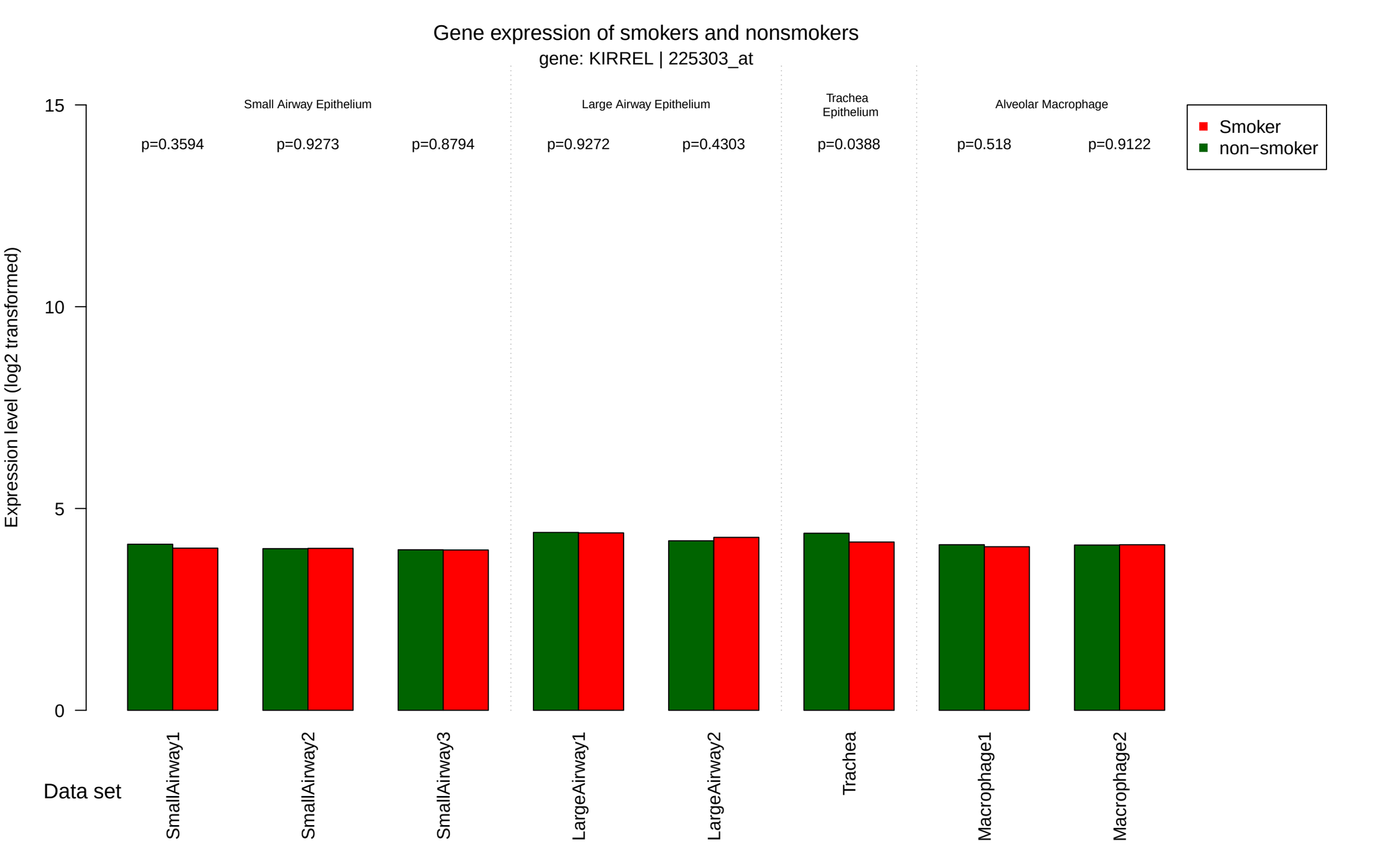
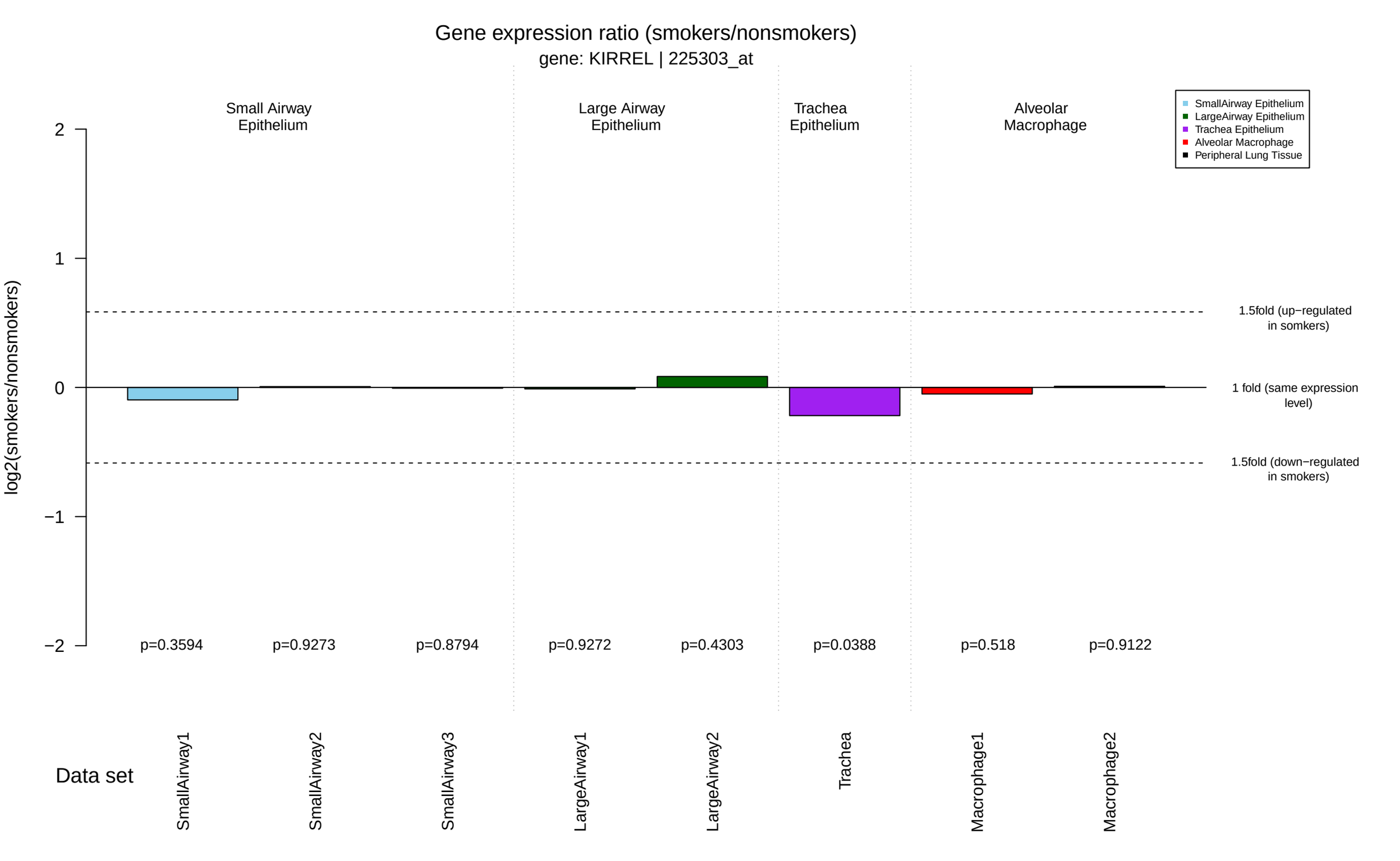
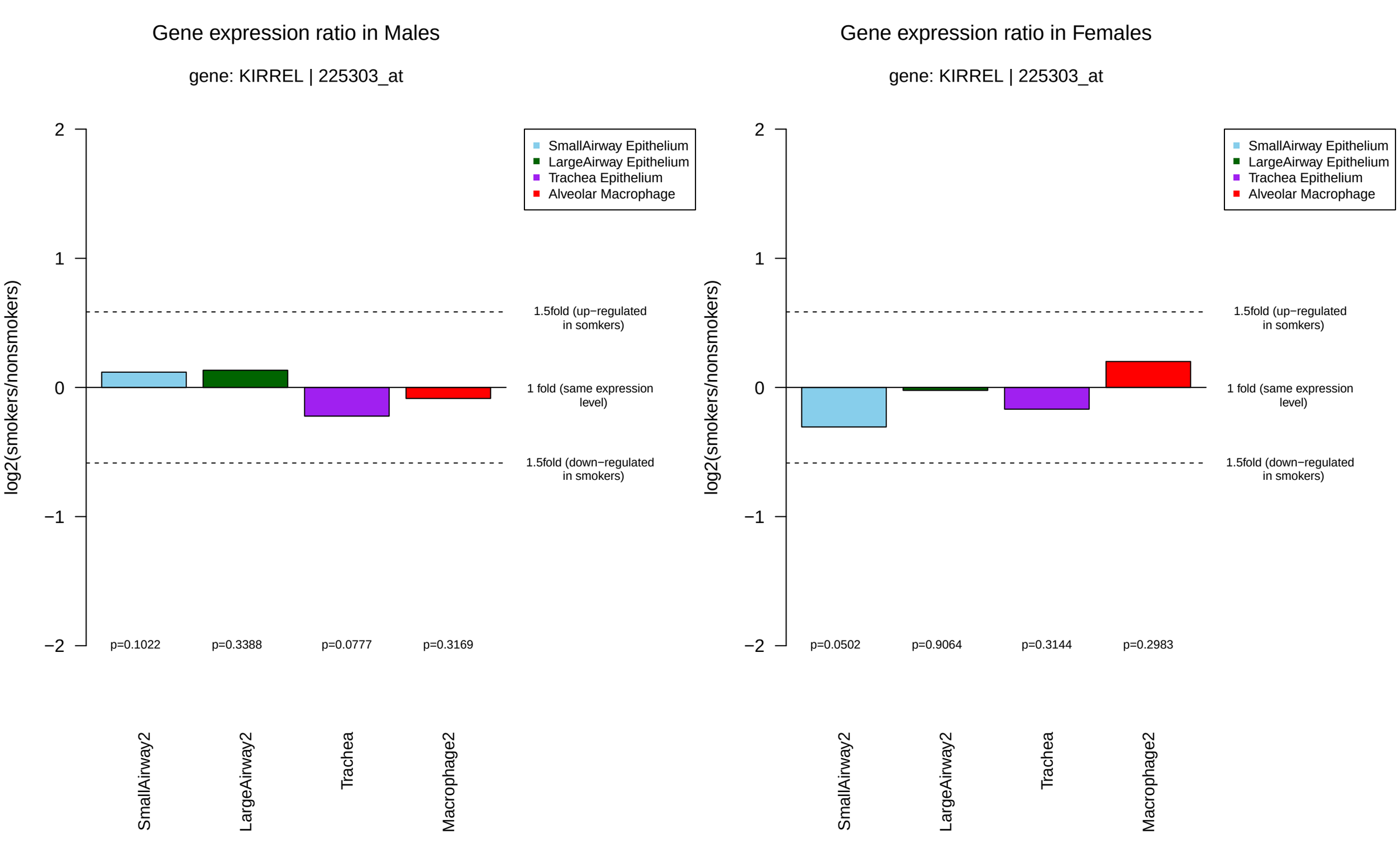
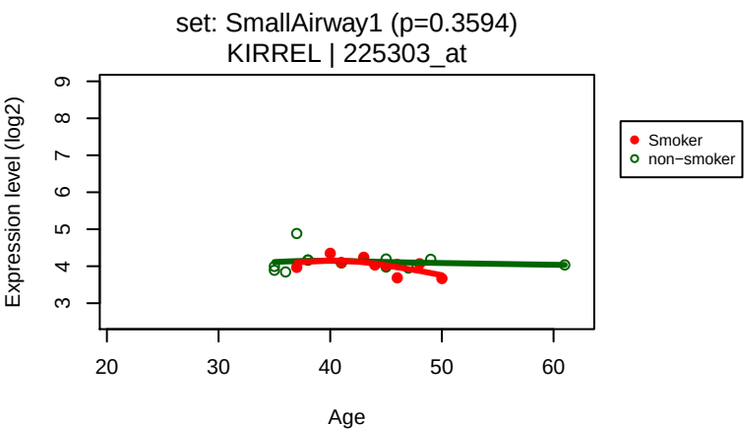
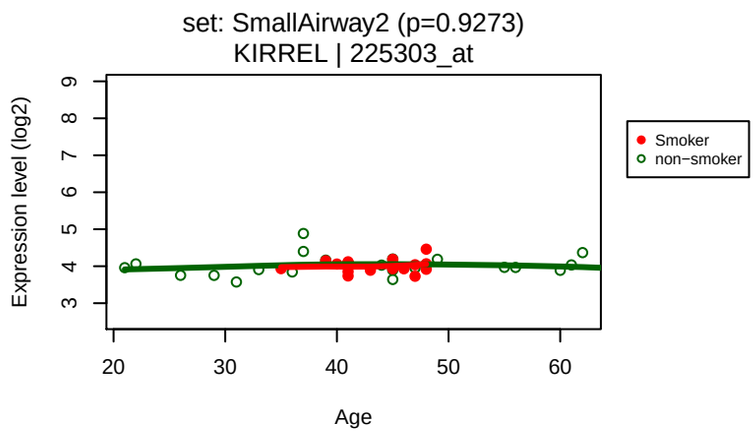
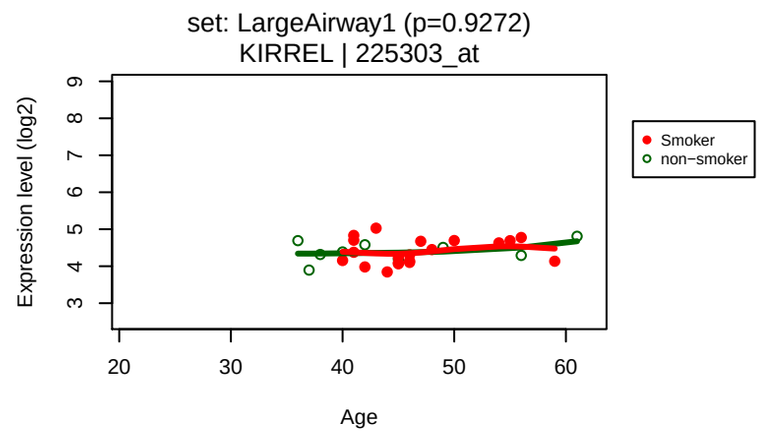

Gene expression of smokers and nonsmokers
gene: KIRREL | 225303_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3594
p=0.9273
p=0.8794
p=0.9272
p=0.4303
p=0.0388
p=0.518
p=0.9122
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

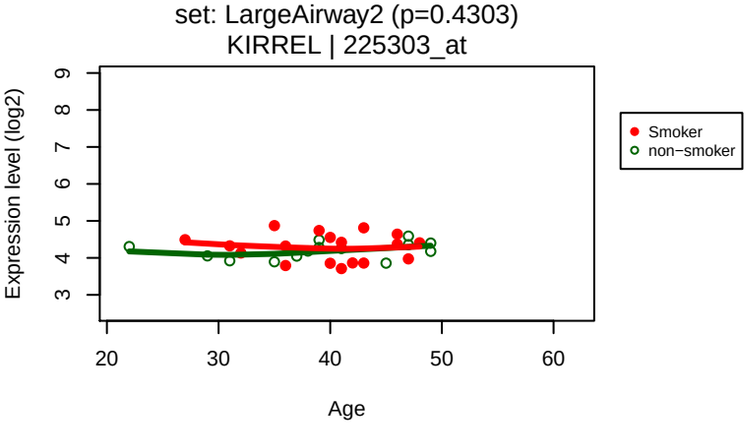

Gene expression ratio (smokers/nonsmokers)
gene: KIRREL | 225303_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3594
p=0.9273
p=0.8794
p=0.9272
p=0.4303
p=0.0388
p=0.518
p=0.9122
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

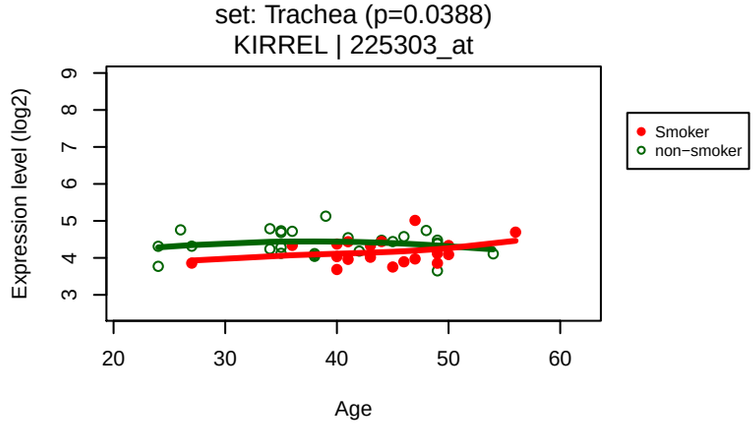

Gene expression ratio in Males
gene: KIRREL | 225303_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1022
p=0.3388
p=0.0777
p=0.3169
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIRREL | 225303_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0502
p=0.9064
p=0.3144
p=0.2983
SmallAirway2
LargeAirway2
Trachea
Macrophage2

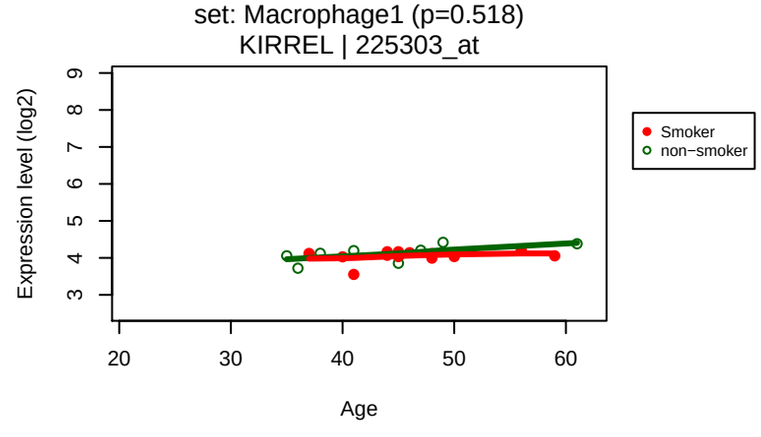

set: SmallAirway1 (p=0.3594)
KIRREL | 225303_at
Expression level (log2)
Age
Smoker
non−smoker

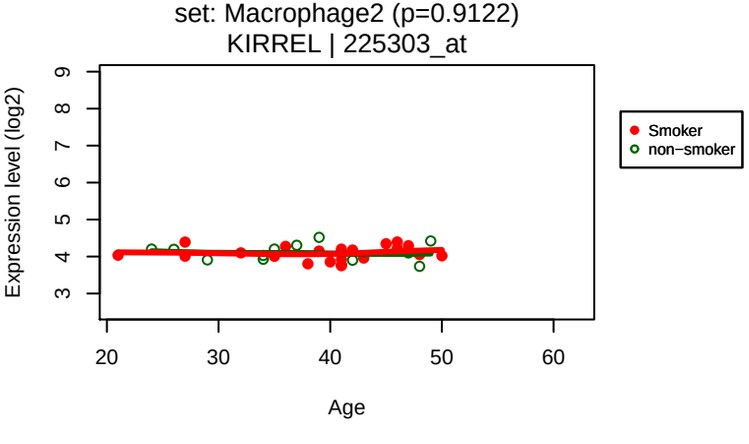

set: SmallAirway2 (p=0.9273)
KIRREL | 225303_at
Expression level (log2)
Age
Smoker
non−smoker

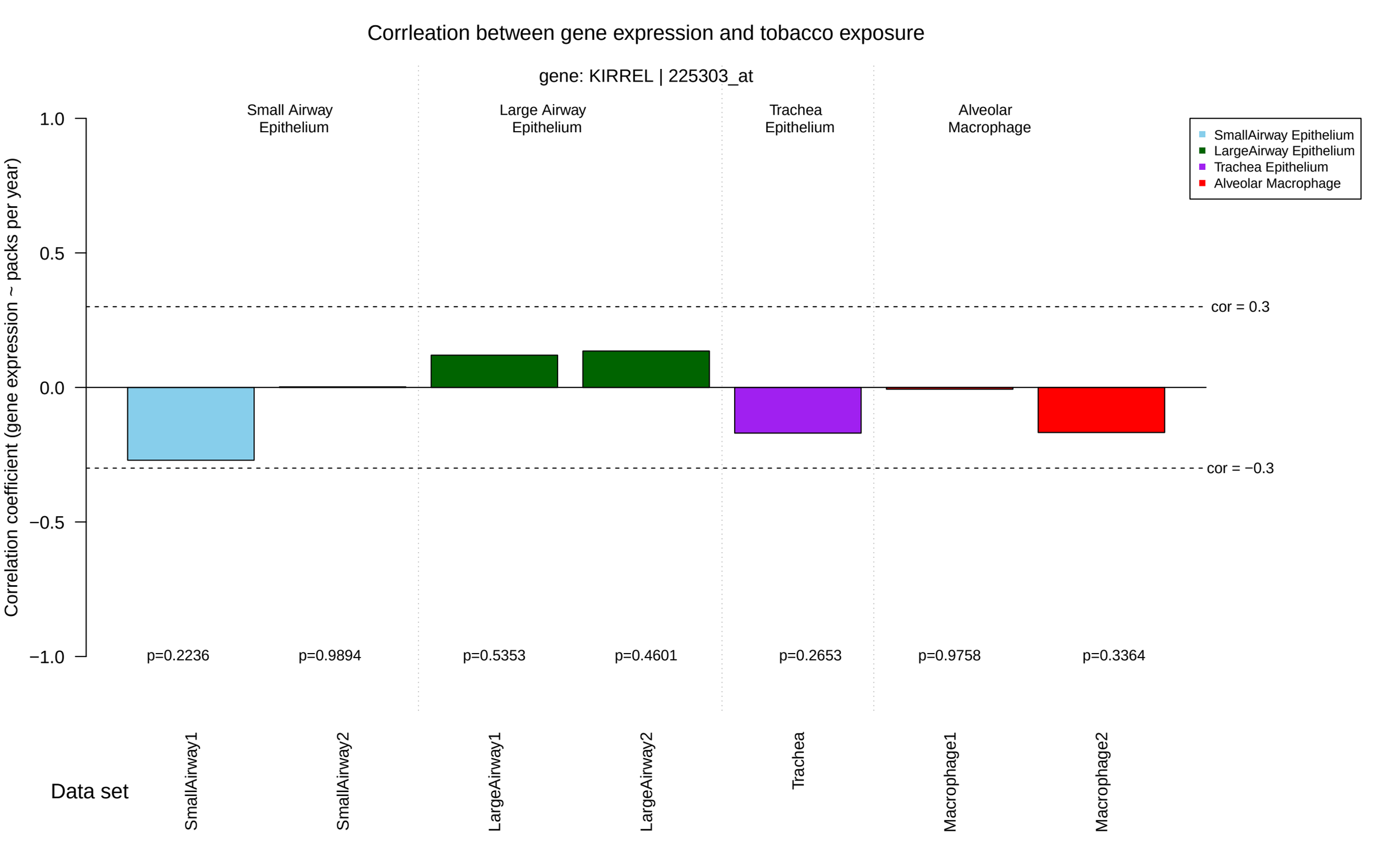

set: LargeAirway1 (p=0.9272)
KIRREL | 225303_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4303)
KIRREL | 225303_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0388)
KIRREL | 225303_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.518)
KIRREL | 225303_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9122)
KIRREL | 225303_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIRREL | 225303_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2236
p=0.9894
p=0.5353
p=0.4601
p=0.2653
p=0.9758
p=0.3364
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2